

University Fest Management - Lab 3 Submission

Name : Bhavani

SRN : PES1UG23CS144

SEMESTER : 5 , SECTION : A

Task 1: Table Creation

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MySQL 8.0 Command Line CLI x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.43 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE University_Fest;
Query OK, 1 row affected (0.08 sec)

mysql> USE University_Fest;
Database changed
mysql> CREATE TABLE Fest ( FestID INT PRIMARY KEY, FestName VARCHAR(100) NOT NULL, Year YEAR NOT NULL, TeamID INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID));
ERROR 1824 (HY000): Failed to open the referenced table 'team'
mysql> CREATE TABLE Team ( TeamID INT PRIMARY KEY, TeamName VARCHAR(100) NOT NULL, MembersCount INT CHECK (MembersCount >= 1), TeamType ENUM('MNG','ORG') DEFAULT 'ORG');
Query OK, 0 rows affected (0.20 sec)

mysql> CREATE TABLE Member ( MemberID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, DOB DATE NOT NULL, Age INT CHECK (Age >= 18), TeamID INT, ReportsTo INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID), FOREIGN KEY (ReportsTo) REFERENCES Member(MemberID));
Query OK, 0 rows affected (0.19 sec)

mysql> CREATE TABLE Event ( EventID INT PRIMARY KEY, EventName VARCHAR(100) NOT NULL, Block VARCHAR(50), Floor INT, RoomNo VARCHAR(20), Date_of_conduction DATE, Price DECIMAL(6,2) CHECK (Price <= 1500), TeamID INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID));
Query OK, 0 rows affected (0.25 sec)

mysql> CREATE TABLE Participant ( SRN VARCHAR(20) PRIMARY KEY, Name VARCHAR(100) NOT NULL, Gender ENUM('M','F','O'), Department VARCHAR(50), Semester INT CHECK (Semester BETWEEN 1 AND 8));
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Visitor ( VisitorID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Gender ENUM('M','F','O'), Age INT CHECK (Age > 0), SRN VARCHAR(20), FOREIGN KEY (SRN) REFERENCES Participant(SRN));
Query OK, 0 rows affected (0.13 sec)
```

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MySQL 8.0 Command Line CLI x + v
ERROR 1824 (HY000): Failed to open the referenced table 'team'
mysql> CREATE TABLE Team ( TeamID INT PRIMARY KEY, TeamName VARCHAR(100) NOT NULL, MembersCount INT CHECK (MembersCount >= 1), TeamType ENUM('MNG','ORG') DEFAULT 'ORG');
Query OK, 0 rows affected (0.20 sec)

mysql> CREATE TABLE Member ( MemberID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, DOB DATE NOT NULL, Age INT CHECK (Age >= 18), TeamID INT, ReportsTo INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID), FOREIGN KEY (ReportsTo) REFERENCES Member(MemberID));
Query OK, 0 rows affected (0.19 sec)

mysql> CREATE TABLE Event ( EventID INT PRIMARY KEY, EventName VARCHAR(100) NOT NULL, Block VARCHAR(50), Floor INT, RoomNo VARCHAR(20), Date_of_conduction DATE, Price DECIMAL(6,2) CHECK (Price <= 1500), TeamID INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID));
Query OK, 0 rows affected (0.25 sec)

mysql> CREATE TABLE Participant ( SRN VARCHAR(20) PRIMARY KEY, Name VARCHAR(100) NOT NULL, Gender ENUM('M','F','O'), Department VARCHAR(50), Semester INT CHECK (Semester BETWEEN 1 AND 8));
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Visitor ( VisitorID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Gender ENUM('M','F','O'), Age INT CHECK (Age > 0), SRN VARCHAR(20), FOREIGN KEY (SRN) REFERENCES Participant(SRN));
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Registration ( RegNo INT, EventID INT, SRN VARCHAR(20), PRIMARY KEY (RegNo, EventID, SRN), FOREIGN KEY (EventID) REFERENCES Event(EventID) ON DELETE CASCADE, FOREIGN KEY (SRN) REFERENCES Participant(SRN));
Query OK, 0 rows affected (0.27 sec)

mysql> CREATE TABLE Stall ( StallID INT PRIMARY KEY, StallName VARCHAR(100) NOT NULL);
Query OK, 0 rows affected (0.31 sec)

mysql> CREATE TABLE Item ( ItemID INT PRIMARY KEY, ItemName VARCHAR(100) NOT NULL, Type ENUM('Veg','Non-Veg'), Price DECIMAL(6,2) NOT NULL DEFAULT 50, MaxStocks INT CHECK (MaxStocks <= 150), StallID INT, FOREIGN KEY (StallID) REFERENCES Stall(StallID));
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Purchase ( PurchaseID INT PRIMARY KEY, SRN VARCHAR(20), ItemID INT, Quantity INT CHECK (Quantity > 0), PurchaseTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (SRN) REFERENCES Participant(SRN), FOREIGN KEY (ItemID) REFERENCES Item(ItemID));
Query OK, 0 rows affected (0.20 sec)

mysql> CREATE TABLE Fest ( FestID INT PRIMARY KEY, FestName VARCHAR(100) NOT NULL, Year YEAR NOT NULL, TeamID INT, FOREIGN KEY (TeamID) REFERENCES Team(TeamID));
Query OK, 0 rows affected (0.39 sec)

mysql> |
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mysql> desc Fest;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FestID     | int           | NO   | PRI | NULL    |       |
| FestName   | varchar(100)  | NO   |     | NULL    |       |
| Year       | year          | NO   |     | NULL    |       |
| TeamID     | int           | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> desc Member;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| MemberID   | int           | NO   | PRI | NULL    |       |
| Name       | varchar(100)  | NO   |     | NULL    |       |
| DOB        | date          | NO   |     | NULL    |       |
| Age        | int           | YES  |     | NULL    |       |
| TeamID     | int           | YES  | MUL | NULL    |       |
| ReportsTo  | int           | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> desc Event;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EventID    | int           | NO   | PRI | NULL    |       |
| EventName  | varchar(100)  | NO   |     | NULL    |       |
| Block      | varchar(50)   | YES  |     | NULL    |       |
| Floor      | int           | YES  |     | NULL    |       |
| RoomNo     | varchar(20)   | YES  |     | NULL    |       |
| Date_of_conduction | date          | YES  |     | NULL    |       |
| Price      | decimal(6,2)  | YES  |     | NULL    |       |
| TeamID     | int           | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.02 sec)

mysql> desc Participant;
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8 rows in set (0.02 sec)

mysql> desc Participant;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SRN    | varchar(20) | NO | PRI | NULL |  |
| Name   | varchar(100) | NO |  | NULL |  |
| Gender | enum('M','F','O') | YES |  | NULL |  |
| Department | varchar(50) | YES |  | NULL |  |
| Semester | int | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+

5 rows in set (0.01 sec)

mysql> desc visitor;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| VisitorID | int | NO | PRI | NULL |  |
| Name     | varchar(100) | NO |  | NULL |  |
| Gender   | enum('M','F','O') | YES |  | NULL |  |
| Age     | int | YES |  | NULL |  |
| SRN     | varchar(20) | YES | MUL | NULL |  |
+-----+-----+-----+-----+-----+-----+

5 rows in set (0.01 sec)

mysql> desc registration;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | NO | PRI | NULL |  |
| EventID | int | NO | PRI | NULL |  |
| SRN   | varchar(20) | NO | PRI | NULL |  |
+-----+-----+-----+-----+-----+-----+

3 rows in set (0.00 sec)

mysql> desc stall;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StallID | int | NO | PRI | NULL |  |
+-----+-----+-----+-----+-----+-----+
```

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+-----+-----+-----+-----+-----+-----+
| RegNo | int   | NO   | PRI | NULL |      |
| EventID | int  | NO   | PRI | NULL |      |
| SRN    | varchar(20) | NO  | PRI | NULL |      |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> desc stall;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StallID    | int           | NO   | PRI | NULL    |      |
| StallName  | varchar(100)  | NO   |     | NULL    |      |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> desc item;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ItemID     | int           | NO   | PRI | NULL    |      |
| ItemName   | varchar(100)  | NO   |     | NULL    |      |
| Type       | enum('Veg','Non-Veg') | YES |     | NULL    |      |
| Price      | decimal(6,2)  | NO   |     | 50.00   |      |
| MaxStocks  | int           | YES  |     | NULL    |      |
| StallID    | int           | YES  | MUL | NULL    |      |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> desc purchase;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| PurchaseID | int           | NO   | PRI | NULL    |      |
| SRN        | varchar(20)   | YES  | MUL | NULL    |      |
| ItemID     | int           | YES  | MUL | NULL    |      |
| Quantity   | int           | YES  |     | NULL    |      |
| PurchaseTime | timestamp     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

Task 2: Table Modifications

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MySQL 8.0 Command Line CLI  ×  +  -  v

+-----+-----+-----+-----+-----+-----+
| PurchaseTime | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> ALTER TABLE Participant MODIFY Gender ENUM('M','F','O') AFTER Name;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE Item MODIFY Price DECIMAL(6,2) NOT NULL DEFAULT 50;
Query OK, 0 rows affected (0.35 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE Item ADD CONSTRAINT chk_stock CHECK (MaxStocks <= 150);
Query OK, 0 rows affected (0.30 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME TABLE Event TO Event_schedule;
Query OK, 0 rows affected (0.07 sec)

mysql> ALTER TABLE Event_schedule MODIFY Date_of_conduction DATE FIRST;
Query OK, 0 rows affected (0.34 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Participant;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SRN | varchar(20) | NO | PRI | NULL | |
| Name | varchar(100) | NO | | NULL | |
| Gender | enum('M','F','O') | YES | | NULL | |
| Department | varchar(50) | YES | | NULL | |
| Semester | int | YES | | NULL | |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> desc Item;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ItemID | int | NO | PRI | NULL | |
| ItemName | varchar(100) | NO | | NULL | |
```

```

MySQL 8.0 Command Line Cli  x  +  v
mysql> desc Participant;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SRN   | varchar(20) | NO | PRI | NULL |  |
| Name  | varchar(100) | NO |  | NULL |  |
| Gender | enum('M','F','O') | YES |  | NULL |  |
| Department | varchar(50) | YES |  | NULL |  |
| Semester | int | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> desc Item;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ItemID | int | NO | PRI | NULL |  |
| ItemName | varchar(100) | NO |  | NULL |  |
| Type | enum('Veg','Non-Veg') | YES |  | NULL |  |
| Price | decimal(6,2) | NO |  | 50.00 |  |
| MaxStocks | int | YES |  | NULL |  |
| StallID | int | YES | MUL | NULL |  |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> desc Event_schedule;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Date_of_conduction | date | YES |  | NULL |  |
| EventID | int | NO | PRI | NULL |  |
| EventName | varchar(100) | NO |  | NULL |  |
| Block | varchar(50) | YES |  | NULL |  |
| Floor | int | YES |  | NULL |  |
| RoomNo | varchar(20) | YES |  | NULL |  |
| Price | decimal(6,2) | YES |  | NULL |  |
| TeamID | int | YES | MUL | NULL |  |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> |

```

Task 3: Theory Questions & Answers

1. The command `SELECT DATABASE () ;` is used to display the currently selected database.
2. The command `CLS` is used to clear the screen.
3. No, you cannot rename a database directly. You must create a new database and then move the data into it.
4. The command `DROP TABLE table_name;` is used to delete a table completely.
5. The `DROP` command deletes both the table and its structure, whereas the `TRUNCATE` command deletes only the rows but keeps the structure intact.
6. No, only one primary key is allowed in a table, but it can be composite, meaning it may consist of multiple columns.
7. Yes, a foreign key can have a `NULL` value unless it is explicitly defined with the `NOT NULL` constraint.
8. No, a primary key cannot have a `NULL` value because of the `PRIMARY KEY` constraint.
9. The `DESC` command shows details such as Field, Type, Null, Key, Default, and Extra.
10. Yes, you can change the primary key by using the command `ALTER TABLE ... DROP PRIMARY KEY, ADD PRIMARY KEY (new_column);`.